

Mathis DELSART - AI/ML Engineer

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SUMMARY

Machine Learning & AI Engineer designing, building, and deploying production-minded AI systems end-to-end. Experience spans training & parallel experimentation on HPC GPU clusters (SLURM), Docker-containerized cloud deployment (Azure), and full-stack delivery from frontend to scalable, API-driven backends. Passionate about Computer Vision, Natural Language Processing, Generative AI (LLMs & RAG), and Agentic AI.

EDUCATION

MSc IN COMPUTER SCIENCE AND ENGINEERING <i>École Polytechnique de Louvain, UCLouvain</i> Major in Artificial Intelligence Graduated with 17.95/20 (SUMMA CUM LAUDE) Thesis: “Deep Reinforcement Learning for Competitive Agents in MicroRTS: Architecture, Training, and Tournament Evaluation” → 📄 PDF	Sept. 2024 - June 2026 Louvain-la-Neuve, Belgium
BSc IN ENGINEERING <i>École Polytechnique de Louvain, UCLouvain</i> Major in Computer Science & Electrical Engineering Graduated with 16.74/20 (MAGNA CUM LAUDE)	Sept. 2021 - June 2024 Louvain-la-Neuve, Belgium

PUBLICATIONS

M. DELSART , A. MORENVILLE, E. PIETTE, “Combining Spatial and Entity-Based Reasoning for Competitive MicroRTS via U-Net and Transformers,” in <i>IEEE Conference on Games (CoG)</i> , 2026, Accepted	Sept. 2026 Madrid, Spain
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PROFESSIONAL EXPERIENCE

PRIVATE TUTOR Mathematics, Physics & Programming Taught 100+ students over 3+ years in math, physics & programming, sharpening my ability to explain technical concepts clearly. Built and deployed a website (mathisdelsart.github.io/private-tutoring) to grow and manage the activity.	2023 - Present Hainaut, Belgium
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PROJECT EXPERIENCE

SOURCIO — RAG STUDY ASSISTANT Deployed RAG tutor with citation-by-construction and a three-layer refusal guard Full-stack tutor that answers only from a user’s own courses (math-aware PDF extraction via a vision model). Every claim is cited by chapter/page <i>by construction</i> — the model never sees a page number, so it cannot invent one. Three guards refuse off-source questions: a calibrated similarity floor, a grounded prompt, and a deterministic no-citation check. Auto-graded exercises & quizzes (LLM-as-a-judge); offline evaluation harness: threshold calibration, retrieval A/B, faithfulness. Live app: sourcio-tutor.vercel.app — 100% citation rate, 96% refusal accuracy on a 71-case endpoint benchmark.	June - July 2026 Python RAG LangChain LLM evaluation FastAPI Qdrant (vector search) LangFuse Next.js / TypeScript PostgreSQL Docker
DRL AGENT FOR COMPETITIVE MICRORTS Custom policy: U-Net + entity-level Transformer (PPO) UECD policy (U-Net · Entity · CBAM · Deep) trained with PPO via self-play and opponent curricula. Extended the Java↔Python env (20+ feature mechanisms) and built a reproducible tournament framework. Topped a 19-agent tournament at 96.67% WR and 65.7% H2H vs prior DRL winner at $\sim 2.5\times$ less compute. Results, replays & analyses: mathisdelsart.github.io/microrts-drl-uecd-website	Sept. 2025 - June 2026 Python PyTorch Deep Reinforcement Learning PPO Transformers Self-Play SLURM (HPC)
SCALABLE CLOUD-NATIVE E-COMMERCE PLATFORM Distributed microservices with elastic auto-scaling Built a microservices platform with 8 backend services behind an API gateway, featuring JWT authentication and a MapReduce-based recommendation engine. Deployed on Azure VMs with Docker Swarm auto-scaling, Redis job queues, and Azure Blob Storage for product images.	Oct. - Dec. 2025 Microservices Node.js SvelteKit Docker Swarm Azure Redis CouchDB
NLP PIPELINE: CLASSIFICATION, EMBEDDINGS & QUESTION ANSWERING From probabilistic models (<i>Naive Bayes</i>) to transformers (<i>BERT</i>) Built Naive Bayes classification reaching $\sim 80\%$ accuracy on 60k Stack Overflow questions, benchmarked sparse vs dense embeddings (TF-IDF, PPML, Word2Vec), and fine-tuned BERT on SQuAD (Stanford Question Answering Dataset) for extractive reading comprehension. » These four are a curated selection — 30+ more projects spanning <i>Systems Programming & Compilers</i> , <i>Low-Level & Embedded</i> , <i>Computer Vision</i> , and <i>Numerical Computing & Optimization</i> , on github.com/mathisdelsart	Oct. - Dec. 2024 Python PyTorch NLP BERT Transformers Hugging Face Word Embeddings

SKILLS

TECHNICAL SKILLS			
ML / DL	Python, PyTorch, scikit-learn, Hugging Face, NumPy, Pandas	GENAI	RAG, LangChain, LangGraph, LLM evaluation, LangFuse
WEB	React, Next.js, Node.js, FastAPI, Tailwind CSS, TypeScript	DATABASES	PostgreSQL, SQLite, CouchDB, Qdrant, Redis
DEVOPS	Git, CI/CD, Docker, Docker Swarm, Azure, SLURM, Bash	AI TOOLING	GitHub Copilot, Claude Code
SOFT SKILLS			
INTERPERSONAL	Communication, Mentoring, Technical Vulgarization	COLLABORATION	Teamwork, Code Review, Knowledge Sharing
MINDSET	Analytical Rigor, Problem-solving, Continuous Learning	WORK ETHIC	Autonomy, Initiative, Perseverance
ORGANIZATION	Time Management, Prioritization, Adaptability		

LANGUAGES

English Professional Working Proficiency | **French** Native